

There are a total of four problems. You have to solve all the problems.

Problem 1 (CO1): DFA and Regular Languages (15 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$$L_1 = \{w : \text{length of } w \text{ is exactly three}\}$$

$$L_2 = \{w : \text{every even position in } w \text{ is } 0\}$$

$$L_3 = \{w : 01 \text{ appears even number of times in } w \text{ as a substring}\}$$

$$L_4 = L_1 \cap L_2 \cap L_3$$

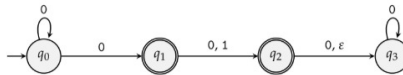
$$L_5 = \{w : 0^m 1^n, \text{ where } m, n \geq 0\}$$

Now solve the following problems.

- Give the state diagram for a DFA that recognizes L_1 . (3 points)
- Give the state diagram for a DFA that recognizes L_2 . (3 points)
- Give the state diagram for a DFA that recognizes L_3 . (3 points)
- If you were to use the "cross product" construction shown in class to obtain a DFA for the language L_4 , how many states would it have? (1 point)
- Find all the strings in L_4 . (1 point)
- Give the state diagram for a DFA that recognizes L_4 using only five states. (2 points)
- Is L_4 a subset of L_5 ? Give justification for your answer. (2 points)

Problem 2 (CO2): Converting NFA to DFA (5 points)

Consider the following NFA:



Now solve the following problems. Note that you do not need to convert the given NFA into its equivalent DFA in order to answer the following questions.

- If you convert the given NFA into an equivalent DFA using the subset construction method, what is the maximum number of states the DFA can have? (1 Point)
- Write the subsets of states of the given NFA that will be the rejecting states in its equivalent DFA (1 Point)
- Write the ϵ -closure of state q_2 in the given NFA. (1 Point)
- What is $\delta(\{q_0, q_3\}, 0)$ in the given NFA? Write all the states. [Recall: $\delta(\{q\}, a)$ is the set of states the NFA transitions to when it is in state q and receives input a .] (2 Points)

Problem 3 (CO1): Regular Expressions (15 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$$L_1 = \{w \text{ does not contain consecutive } 1\}$$

$$L_2 = \{w \text{ starts with } 0\}$$

$$L_3 = \{w \text{ starts and ends with the same character}\}$$

$$L_4 = L_2 \setminus L_3$$

Now solve the following problems.

- Give a regular expression for the language L_1 . (3 points)
- Give a regular expression for the language L_2 . [Recall: $\overline{L_2}$ denotes the complement of the language L_2 i.e., $\overline{L_2} = \Sigma^* - L_2$] (3 points)
- Give a regular expression for the language L_3 . (3 points)
- Write four four-letter strings in L_4 . (2 point)
- Give a regular expression for the language L_4 . [Recall: $L_2 \setminus L_3$ contains all strings that are in L_2 but not in L_3] (2 points)
- Give a regular expression for the language $\overline{L_4}$. (2 points)

Problem 4 (CO2): Converting Regular Expressions to NFA (10 points)

Convert the following regular expression over $\Sigma = \{a, b\}$ into an equivalent NFA. Note that $R_1 + R_2$ is the same as $R_1 \cup R_2$.

$$(ba)^*a + bb(a + a^*b)^*$$

After you are done with the test, please indicate where you stand on the smiley face spectrum.

